Group Bulk Ordering App: Requirements Document

1. Purpose and Scope

This document defines the requirements for a mobile-first app that helps a group organizer coordinate bulk grocery and household purchases, with other people joining the group to split the cost and the product.

Requirements are derived from interview findings and are tagged **Must**, **Should**, or **Could** for prioritization.

Target users (initial): University students in Victoria, BC (UVic area), most of whom shop at Walmart or Costco and many of whom live with roommates. Interviewees showed a bit of hesitation in respect to having complete strangers shop for them. Students have **larger community groups** (roommates, clubs, dormmates, teams) and showed an **emphasis on convenience** because of their busy schedules.

Out of scope for Milestone 2: Direct in-app payment processing, focus should be on the group creation/order joining process

2. Key Findings from Interviews

2.1 Current shopping habits

- Most shop **weekly** (often on a set day), a few every two weeks or monthly; one shops several times a week.
- **Walmart and Costco** dominate. Staples are milk, eggs, chicken, rice, fruit, and cereal.
- Typical **spend** ranges from about $50–$**100 per week** up to $150–$**300 per month**; a **trip takes about an hour** including travel (range: 20 minutes to 2 hours).
- Package size is driven by **spoilage and consumption**, not just price (I2, I7, I8, I9). Several people have switched from gallon milk to smaller cartons for this reason.
- Cost-saving tactics are mostly informal: store brands, value sizes, buses instead of rideshares, a delivery pass (I7).

2.2 Bulk buying

- A majority already buy some items in bulk: **meat/chicken, rice, eggs, toiletries, and other non-perishables**.
- **Why they buy in bulk:** lower cost (the most common answer), fewer restocking trips, and splitting with roommates.

- **Barriers:** upfront cost (I5, I7: "money gives anxiety"), limited fridge/freezer space (I8), spoilage of perishables like milk and produce (I9, I10), nobody to split with (I4), and no car or Costco membership.
- **Unmet demand:** larger milk sizes and produce (I1), larger egg trays (I2), meat if freezer space allowed (I8), and smaller-share vegetable packs (I11).

2.3 Sharing purchases today

- Most interviewees have roommates. The common pattern is **shared staples, separate everything else**, with coordination through a **group chat** and settlement via **e-transfer**, usually by splitting a receipt (I1, I3, I9, I11).
- Tracking is weak: some rely on memory and admit to forgetting payments (I1), one uses Splitwise (I3), one uses an honor system ("if you use it, refill it", I6).
- Logistics create real friction: one household has 4 roommates, 2 cars, and 3 Costco memberships (I6), and sorting out whose card and car to use is a recurring task (I7). Gas is sometimes split (I3, I6).

2.4 Trust in strangers

- Openness is **conditional but real**. Several would join a group with strangers (I1, I5, I6, I7, I8, I10); others only with known people (I2, I11, I4 and I9 are maybe/roommates only).
- Conditions named: a **safe, public pickup spot** (I1), a pickup within a **10-minute walk or on campus** like UVic (I5), **real savings** (I3), **someone handling portioning** to prevent waste (I7), good **produce quality** (I7), and a smaller share size that solves storage (I9).
- One person already trusts a stranger to shop for her through delivery (I8), which suggests the concept is not foreign.

2.5 Organizer experience (the critical risk)

- **Nobody interviewed had organized a group purchase beyond informal roommate trips.**
- Willingness to organize is low: most would use the app but **not as an organizer**. Only one interviewee was open to organizing occasionally, and several of those who might organize would do so for friends or family only, with a maximum group size of roughly **3 to 10** (typically 5–7).
- Reasons for declining: no car or space (I4, I7), already time-consuming own shopping (I8).
- Most time-consuming organizer tasks: **finding a time that works for everyone and the drive** (I11), **finding unfamiliar items** (I3), and **deciding whose card and car to use** (I7).
- Some interviewees said no one needed to be compensated because trips are casual (I6), so the incentive to organize is currently undefined.

Implication: The app succeeds or fails on how little work it asks of the organizer and whether the organizer gets something in return. A large pool of willing joiners with too few organizers will leave the product empty.

3. User Roles

- **Organizer:** Creates a group order, sets items and deadlines, purchases the goods, and hands off items. Needs a car and ideally a membership (e.g., Costco).
- **Member:** Joins a group, commits to a quantity, pays their share, and collects their items.

4. Functional Requirements

4.1 Accounts and trust

- **FR-1 (Must):** Users sign up with a verified email. A university email (e.g., uvic.ca) badge is displayed on profiles to build trust. *(Basis: stranger trust conditions)*
- **FR-2 (Must):** Profiles show display name, photo, and number of completed orders
- **FR-3 (Should):** Organizers can mark a group as "friends only" (invite link) or "open to community" (discoverable)
- **FR-4 (Should):** Members can rate organizers and organizers can rate members after each order.
- **FR-5 (Should):** Users can report a member or organizer, and organizers can remove members from a group.

4.2 Group creation and discovery

- **FR-6 (Must):** An organizer can create a group order with: store (e.g., Costco, Walmart), order deadline, purchase date, pickup location and time, and maximum group size.
- **FR-7 (Must):** Default and recommended maximum group size is **6**, configurable up to **10**.
- **FR-8 (Must):** Members join through an invite link or by browsing open groups near them.
- **FR-9 (Should):** Members can browse open groups filtered by store, pickup location, and date.
- **FR-10 (Could):** Recurring groups (weekly or monthly) that auto-open each cycle, since many people shop on a fixed day.

4.3 Items, quantities, and splitting

- **FR-11 (Must):** The organizer adds items with name, store price, pack size, and total pack count available. Members request a quantity of a **share** (e.g., 1/3 of a chicken tray, 6 eggs out of 30).

- **FR-12 (Must):** Items support **splittable shares**, so a pack that is too big for one person (egg trays, meat, produce, milk) can be divided among members. *(Basis: I1, I2, I8, I11 unmet demand)*
- **FR-13 (Must):** The app shows each member's running cost
- **FR-14 (Should):** Items are labeled with a **perishability** indicator and recommended storage (freezer-friendly, shelf life), and the app warns when a share may spoil before use. *(Basis: spoilage and storage concerns from I7, I8, I9, I10)*
- **FR-15 (Should):** The organizer can mark portioning responsibility, and the app lists portioning steps for the pickup. *(Basis: I7)*
- **FR-16 (Could):** Item library of common staples (milk, eggs, chicken, rice, tissues, toilet paper, cereal) with typical pack sizes to speed up order creation.

4.4 Costs and payments

- **FR-17 (Must):** The app calculates each member's exact total automatically, including taxes and any shared fees, so no one needs to split a receipt manually.
- **FR-18 (Must):** Members can see the amount owed, mark it as paid (e-transfer is the expected method), and the organizer can confirm the payment was received.
- **FR-19 (Must):** Dashboard for the organizer showing who has committed, who has paid, and what is outstanding.
- **FR-20 (Should):** Optional **organizer compensation**: a gas/travel share or a small service fee, split transparently across members. *(Basis: organizer time and transport costs are unrecognized today)*
- **FR-21 (Should):** Option to **collect payment before purchase** (to reduce the organizer's upfront cost risk), with the final receipt reconciled afterward.
- **FR-22 (Could):** After shopping, the organizer uploads the receipt and the app adjusts shares for price differences or unavailable items.
- **FR-23 (Could):** Export or link to Splitwise-style balances for housemates.

4.5 Changes after commitment

- **FR-25 (Must):** If an item is out of stock or the price changes, the organizer can update the order and the app notifies all affected members and recalculates totals.
- **FR-26 (Should):** A minimum-participation threshold: the order only proceeds if enough shares are committed, otherwise it is cancelled automatically.

4.6 Pickup and logistics

- **FR-27 (Must):** The organizer sets a pickup location and time slot; members confirm they can attend.
- **FR-28 (Should):** Suggested **safe public pickup points**, including UVic campus locations, and a note of walking time from the member's location. *(Basis: I1, I5)*
- **FR-30 (Could):** Pickup check-in so the organizer can mark that a member has picked up their items.

4.7 Communication

- **FR-31 (Must):** Push or email notifications for: new group invites, deadline reminders, price or item changes, payment requests, and pickup reminders.

4.8 Quality of Life features

- **FR-34 (Could):** One-tap reorder of a past group order, inspired by how quickly people reorder on Instacart (I5, ~3 minutes).
- **FR-35 (Could):** Estimated savings display comparing the group price per unit with a typical smaller-pack price. *(Basis: I3 needs to know how much they are actually saving)*

5. Non-Functional Requirements

- **NFR-1 (Must):** Mobile-first design; the typical join-and-commit flow should take **under 2 minutes**.
- **NFR-2 (Must):** The organizer flow should take **under 10 minutes** to set up a standard order, since time is the main reason people decline to organize.
- **NFR-3 (Must):** Personal data (names, contact details, locations) is stored securely and shown only to members of the same group; exact addresses are never shared by default.
- **NFR-4 (Should):** Works without requiring users to install anything beyond the app; sign-in with a university email.
- **NFR-5 (Should):** Clear, consistent display of money in CAD with taxes shown.
- **NFR-6 (Could):** Accessible design following WCAG AA.

6. Risks and Mitigations

- **Too few organizers.** Mitigate with low-effort setup (FR-6, FR-11, FR-16), organizer compensation (FR-20), prepaid collection (FR-21), and recurring groups (FR-10).
- **Trust between strangers.** Mitigate with verified university emails, ratings, reporting, public pickup, and "friends only" groups (FR-1 to FR-5, FR-28).
- **Food waste or spoilage.** Mitigate with shares sized to individual use, perishability warnings, and portioning support (FR-12, FR-14, FR-15).
- **Payment disputes.** Mitigate with automatic cost calculation, paid/confirmed states, and receipt reconciliation (FR-17, FR-18, FR-22).
- **The organizer is paying out of pocket.** Mitigate with pre-purchase collection and minimum thresholds (FR-21, FR-26).